

Risk Prediction and Interpretation for Fall Events Using Explainable AI and Large Language Models

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Abstract

Falls represent a significant public health concern, particularly among older adults, with approximately 35.6 million cases annually in the United States alone. Traditional machine learning-based fall risk prediction methods often lack precision and interpretability, limiting their effectiveness in clinical settings. This study aims to develop and validate an explainable machine learning approach for fall risk prediction that combines predictive accuracy with interpretable results and leverages large language models for enhanced communication, enabling healthcare providers to make informed decisions about fall prevention strategies. We developed an integrated machine learning pipeline utilizing XGBoost classifier to process key health indicators such as age, and diagnosis history, with the model's interpretability enhanced through SHAP (SHapley Additive exPlanations) values. Results were transformed into natural language narratives using LangChain and large language models, which automated the generation of personalized reports combining both risk assessments and feature explanations, with patients classified into low (0-40%), medium (40-70%), and high (70-100%) risk groups based on predicted probabilities. The XGBoost classifier emerged as the best-performing model, achieving 71% accuracy, 69% precision, 76% recall, an F1-score of 72%, and an ROC AUC of 0.71 on the testing dataset. The SHAP analysis provided transparent insights into feature importance, highlighting critical contributors to fall risk prediction. The integration of large language models through LangChain successfully transformed

complex model outputs into comprehensible narratives, generating detailed explanations that included both high-level risk summaries and specific feature contributions for healthcare providers and patients. Our integrated approach demonstrates the feasibility of combining high-accuracy fall risk prediction with explainable AI techniques and large language models, with the system's ability to provide interpretable results and generate clear, personalized risk communications representing a significant advancement in developing practical AI-driven tools for clinical fall risk assessment, potentially improving the implementation of preventive interventions in healthcare settings.

CCS Concepts

• Human-centered computing; • Human computer interaction (HCI); • Empirical studies in HCI;

Keywords

Fall Risk Prediction, Explainable Artificial Intelligence, Clinical Decision Support Systems, Large Language Models, Machine Learning in Healthcare, SHAP Values

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1 Introduction

Falls are the leading cause of injury-related death among US adults aged 65 and above, posing a significant public health concern [1, 2]. The scale of this problem is substantial: each year, approximately 35.6 million people experience falls in the United States alone, with 8.4 million resulting in reported injuries. The economic burden is

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equally significant, with each fall-related hospitalization costing an average of \$14,056, contributing to total annual medical costs exceeding \$50 billion [1, 3]. The global impact of falls is equally concerning. Falls account for over 38 million disability-adjusted life years lost annually, surpassing the combined impact of transport injuries, drownings, burns, and poisonings [4]. Among older adults, more than one out of four individuals fall each year, with one in five falls resulting in serious injuries such as broken bones or head trauma.

Research indicates that early assessment and appropriate intervention could prevent nearly one-third of falls [5]. This early intervention is particularly crucial, as studies have shown that many older patients experience recurrent falls within 6-12 months after their first medically attended fall event. Moreover, the initial fall often leads to decreased fall self-efficacy (FSE), which has been identified as a significant predictor for recurrent falls. However, despite recommendations for universal fall risk screening, the actual implementation of preventative interventions remains low [6, 7]. The implementation gap can be attributed to several factors, including the lack of interpretable and accurate risk prediction tools that can provide actionable insights for healthcare providers.

This study addresses these challenges by developing a novel explainable machine learning system that combines high-accuracy fall risk prediction with comprehensive interpretability through advanced explainable artificial intelligence (XAI) techniques. Our approach integrates state-of-the-art machine learning models with XAI-based feature analysis and large language model-powered explanations, providing healthcare providers with both accurate risk assessments and clear, actionable insights into the factors driving these predictions. This integration of explainable AI techniques is vital for enabling informed clinical decision-making and developing targeted intervention strategies, ultimately improving patient outcomes and reducing the burden of falls in healthcare settings.

2 Related Studies

2.1 Machine learning in health care and fall prediction

Machine learning has been used in healthcare prediction and decision support across numerous domains, demonstrating remarkable success in early diagnosis, risk assessment, and treatment planning. In clinical settings, ML models have achieved significant breakthroughs in predicting hospital readmissions with accuracy rates exceeding 80% [8], forecasting disease progression in conditions such as Alzheimer's [9], stroke [10], and optimizing patient care pathways through automated triage systems [11]. ML models have been employed to predict 30-day hospital readmissions [12] with high accuracy by analyzing patient demographics, comorbidities, and prior hospital visits. Similarly, in chronic disease management, ML algorithms have been used to identify patients at risk of developing hypertension [13], diabetes [14], or arthritis [15] conditions, enabling earlier interventions. These applications illustrate ML's potential to transform healthcare by providing actionable insights that support timely and informed decision-making.

In the domain of fall risk assessment, researchers have developed various ML-based approaches, primarily focusing on post-fall

prediction and recurrent fall prevention. Recent studies have utilized sensor-based data collection methods [16], with Chen et al. achieving notable success using wireless sensor-embedded systems combined with SVM [17]. Other approaches have incorporated accelerometer and gyroscope data with support vector machines, demonstrating good accuracies in controlled settings [16]. However, these methods typically rely on physical movement data or post-fall information, limiting their ability to predict falls before they occur.

Our approach differentiates itself in several aspects. First, we leverage comprehensive electronic health records (EHRs) [18] data rather than sensor-based measurements, enabling us to predict falls before they occur rather than focusing on recurrent fall prevention. Second, our model incorporates a broader range of clinical variables, including demographics and comorbidities, providing a more holistic assessment of fall risk factors. Most importantly, we address the critical need for interpretability in clinical decision support by integrating advanced explainable AI techniques, which not only predict fall risk but also provide healthcare providers with clear, actionable insights into the factors driving these predictions. This emphasis on explainability through sophisticated XAI methods represents a significant advancement in making ML-based fall prediction both accurate and clinically meaningful.

2.2 Explainable AI (XAI) Techniques

Explainable AI (XAI) has become increasingly crucial in healthcare applications, where understanding model decisions is as important as the predictions themselves [19, 20]. This need for transparency is particularly critical in clinical settings, where decisions can have life-altering consequences and providers trusted choices. Traditional "black box" ML, despite their high accuracy, often fail to meet healthcare providers' needs for clear, justifiable decision-making processes, potentially limiting their practical implementation in clinical settings.

Various XAI techniques have been developed and applied in healthcare, each offering different approaches to model interpretation. These tools have been used in many domains for providing clear, detailed insights into complex predictive models, particularly in deep learning frameworks. For example, Local Interpretable Model-agnostic Explanations (LIME) has been successfully employed in clinical settings, helping clinicians understand why ML models make specific diagnostic recommendations [21]. Feature importance methods, such as SHAP (SHapley Additive exPlanations) values, have been particularly effective in clinical prediction tasks, providing insights into how different patient characteristics influence model predictions [22, 23].

While XAI techniques provide valuable insights into model decisions, the technical nature of their outputs - such as feature importance scores or SHAP values - can still present barriers to practical clinical implementation. Recent advances in Large Language Models (LLMs) [24] offer a promising solution to this challenge by transforming complex XAI outputs into natural, contextual explanations that healthcare providers can readily understand and act upon [25]. This integration of LLMs with XAI represents a significant advancement in making machine learning models truly interpretable in

clinical settings. Traditional XAI approaches often present information in formats that require additional interpretation - for instance, showing that a particular lab value contributed 23% to a prediction, or that medication history increased risk by a certain factor. While mathematically precise, such explanations may not immediately translate into actionable clinical insights. LLMs can bridge this gap by generating narrative explanations that place these technical insights within the broader context of a patient's clinical history and current condition [26]. Rather than simply highlighting the importance of certain features, an LLM can explain how a patient's recent medication changes, combined with their existing conditions and mobility status, collectively contribute to their elevated fall risk.

In our approach, we leverage LLMs to enhance XAI in three ways. First, we translate SHAP values and other feature importance metrics into coherent clinical narratives that explain not just what factors contribute to fall risk, but why and how they interact. We generate personalized recommendations that account for both the quantitative risk factors and the qualitative aspects of patient care. We also provide explanations at different levels of detail, allowing healthcare providers to quickly grasp key insights while having access to more detailed explanations when needed. This multi-layered approach to explainability makes our system's predictions more transparent, trustworthy, and actionable in clinical practice.

3 Methods

We integrate three key components to create an interpretable and accurate fall risk prediction system: (1) a machine learning pipeline for fall risk prediction using electronic health records (EHR), (2) an explainable AI framework based on SHAP values for model interpretation, and (3) a large language model-based system for generating natural language explanations. Figure 1 provides an overview of our system pipeline.

The fall risk prediction component compares machine learning algorithms and selects Random Forest, an ensemble learning method chosen for its superior performance with heterogeneous medical data and its robust handling of feature interactions and variability. Our model processes comprehensive EHR data, including demographics and medical history, to generate individualized fall risk predictions.

To address the critical need for model interpretability in healthcare, we implement a multi-layered explainable AI framework using SHAP values [22, 23]. This framework provides three levels of interpretation: (1) individual prediction explanations detailing how specific health indicators contribute to a patient's risk score.

The final component of our system transforms technical model outputs into clear, actionable insights through large language model-based natural language generation. This component takes the SHAP values and risk predictions as input and generates comprehensive narrative explanations that contextualize the risk factors within each patient's unique clinical profile. These explanations are designed to be both technically accurate and clinically meaningful, facilitating better communication between healthcare providers and enabling more informed decision-making about fall prevention strategies.

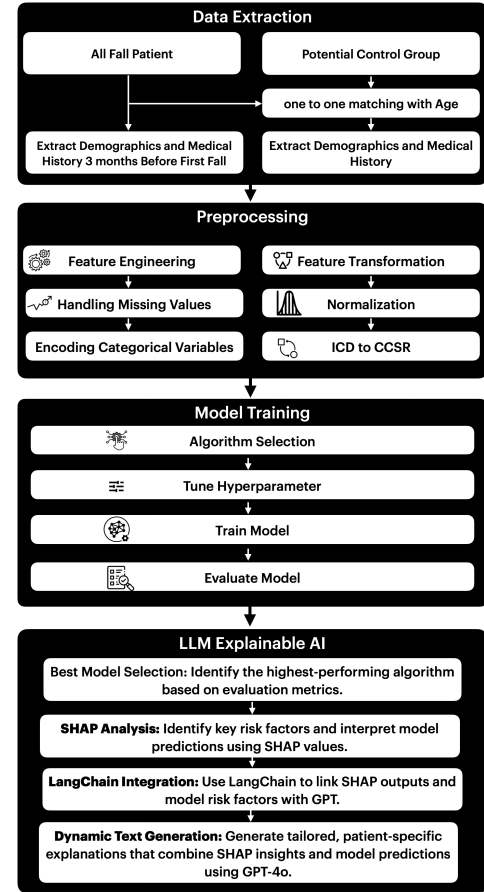


Figure 1: Overview of the fall risk prediction pipeline. The process begins with data extraction, followed by preprocessing, model training, and evaluation. The final step leverages LLM-based Explainable AI, integrating SHAP values and model-derived risk factors through LangChain and GPT-4 to generate dynamic, patient-specific explanations.

3.1 Data Source and Data Processing

Our study utilized electronic health records (EHR) [18] from the Froedtert Hospital system collected between 2015 and 2024. The dataset comprised 198,156 patients in total, including 99,078 patients who experienced falls, and 99,078 one-to-one matched patients based on age without fall incidents as controls. To ensure our model predicts future fall risk rather than detecting existing falls, all features were extracted from data collected three months prior to any recorded fall events for the fall group, and from randomly selected time points for the control group.

The extracted features encompassed multiple categories of health indicators known to influence fall risk including: 1) Demographics: age, gender, race, ethnicity; 2) Medical history: diagnoses, procedures. Data preprocessing involved several key steps. First, missing numerical values were imputed using the mean of the available values to maintain data consistency and minimize bias. Second, numerical features were normalized to a 0-to-1 range to ensure

that all measurements contributed proportionately and to stabilize model training. Third, categorical variables were transformed using one-hot encoding, converting each category into a binary indicator feature to capture discrete patient attributes. Fourth, International Classification of Diseases (ICD) codes were aggregated into Clinical Classifications Software Refined (CCSR) categories, reducing dimensionality and improving interpretability by grouping related diagnoses into broader clinical concepts.

To ensure robust model development and evaluation, we implemented a systematic data splitting strategy. The complete dataset was divided into training (80%) and testing (20%) sets, with stratification to maintain consistent fall-to-control ratios across both sets. This splitting ratio provided sufficient data for model training while ensuring a representative testing set for reliable performance assessment.

To identify the most effective approach for fall risk prediction, we conducted a comprehensive evaluation of multiple machine learning algorithms. We began with a baseline logistic regression model to establish fundamental performance metrics. From there, we explored a diverse set of advanced techniques, including decision trees for interpretable rule-based classification, random forests and extra trees for their ensemble learning robustness, gradient boosting for its ability to iteratively refine weak learners, AdaBoost for adaptive boosting, and XGBoost for its efficiency and advanced gradient boosting framework. By comparing the predictive capabilities of these models, we aimed to identify the most effective algorithm for accurately forecasting fall risk.

The comparative analysis employed a rigorous 5-fold cross-validation methodology on the training dataset. Performance evaluation encompassed multiple complementary metrics to ensure a comprehensive assessment of each algorithm's capabilities. The primary metrics included the Area Under the ROC Curve (AUC) for discrimination ability, accuracy to measure overall correct predictions, precision and recall to evaluate predictive accuracy for positive cases, the F1-score to balance precision and recall, and sensitivity and specificity to assess the algorithm's capability to correctly identify both fall and non-fall cases.

Following model selection, we implemented a comprehensive hyperparameter optimization strategy using grid search. This process systematically adjusted critical parameters, including the regularization strength and solver for logistic regression, maximum tree depth and splitting criteria for decision trees, and the number of estimators, learning rate, and subsample ratio for ensemble methods such as Random Forest, Gradient Boosting, AdaBoost, Extra Trees, and XGBoost.

The optimization process involved exploring parameter combinations within predefined ranges using 5-fold cross-validation, ensuring robust evaluation across multiple data splits. This thorough approach enabled us to identify the most effective configurations for each model, achieving an optimal balance between predictive accuracy and generalizability. This rigorous model selection and optimization process yielded a robust predictive framework capable of accurate fall risk assessment across diverse patient populations. The final model's performance is particularly noteworthy given the inherent complexity of fall risk prediction in healthcare settings, where multiple interacting factors contribute to patient outcomes.

3.2 Explainable AI Modeling

To address the critical need for transparency in healthcare decision support systems, we implemented SHAP (SHapley Additive exPlanations) as our primary model interpretation framework. This approach enabled us to quantify the contribution of each feature to the model's predictions, providing healthcare providers with clear insights into the factors influencing fall risk assessments.

For each prediction, SHAP values were calculated by simulating the presence and absence of features in the model and measuring their marginal contributions. These values were visualized using techniques such as force plots for individual explanations and summary plots for identifying global trends. These visualizations provided intuitive representations of how features interact to influence predictions, making the insights accessible to healthcare providers with varying technical expertise.

The SHAP framework was particularly valuable in generating patient-specific explanations, allowing us to understand how each feature either increased or decreased the fall risk for individual cases. This granular level of interpretation was essential for healthcare providers to make informed decisions based on the model's predictions. The analysis revealed complex interactions between different health indicators and their impact on fall risk, providing valuable insights for clinical practice.

To make the risk assessments more actionable, we developed a comprehensive risk categorization system. The system classified patients into three risk levels based on their predicted probabilities: low risk (0-40%), medium risk (40-70%), and high risk (70-100%). This stratification provided healthcare providers with clear thresholds for decision-making while maintaining the nuanced probability information from the model. Each risk category was accompanied by detailed SHAP-based explanations, ensuring that the categorization was both interpretable and clinically meaningful.

3.3 Interpret XAI Output with Large Language Models

To bridge the gap between technical model outputs and clinical utility, we integrated Langchain [27] to generate comprehensive, natural language narratives explaining the model's predictions. Langchain is an application framework to accelerate LLM application development. This integration enabled us to transform complex SHAP values and risk assessments into clear, contextual explanations that could be readily understood by both healthcare providers and patients. The process began with calculating SHAP values for individual patients to identify the features most impactful to their fall risk. These values were sorted in descending order of magnitude to prioritize the most critical contributors. For instance, if a patient's age had the largest influence on their prediction, these factors were highlighted in the narrative, along with their direction of impact (positive or negative).

We used LangChain to build an agent capable of transforming these insights into comprehensive narratives. Structured prompt templates formed the core of this system, combining a high-level risk summary with detailed SHAP feature explanations. The templates included specific instructions, such as summarizing the patient's overall risk as low, medium, or high, and explaining the top contributing factors with their patient-specific values. LangChain

Table 1: Performance Comparison of Different Models for Three-Month Fall Risk Prediction

Model	Accuracy	Precision	Recall	F1-Score	Sensitivity	Specificity
Logistic Regression	0.63	0.64	0.62	0.63	0.62	0.65
Decision Tree	0.63	0.69	0.51	0.58	0.51	0.77
Random Forest	0.71	0.70	0.74	0.72	0.74	0.68
Gradient Boosting	0.61	0.69	0.70	0.70	0.70	0.68
Extr Trees	0.69	0.68	0.75	0.71	0.75	0.64
XGBoost	0.71	0.69	0.76	0.72	0.76	0.66

enabled the seamless interaction between the SHAP-derived insights and GPT-4, dynamically generating clear and personalized explanations for each patient.

This approach ensured that healthcare providers received both a concise risk summary and detailed explanations of the underlying drivers, organized hierarchically for ease of understanding. The system was also designed to adapt its output based on the target audience. For clinicians, the narratives included technical details and quantitative insights, while for patients, the focus was on actionable recommendations and simplified language. By integrating SHAP transparency with GPT-4’s natural language generation capabilities, LangChain allowed us to create a bridge between AI model predictions and real-world clinical decision-making. This combination of interpretability and usability empowered healthcare providers to make informed decisions while enhancing trust in the system’s outputs.

4 Experiments and Results

4.1 Machine Learning Evaluation

The XGBoost-based fall risk prediction model demonstrated robust performance in experimental evaluation (Table 1). The model achieved a classification accuracy of 71% on the testing dataset, which represents a significant improvement over traditional risk assessment methods. This performance level is particularly meaningful in the context of fall risk prediction, where the complexity of contributing factors often makes accurate prediction challenging.

The results in Table 1 demonstrate that modern machine learning approaches can achieve significant predictive accuracy, with the XGBoost classifier achieving 71% accuracy, 69% precision, 76% recall, and an F1-score of 72% in fall risk prediction. Additionally, the model achieved a ROC AUC of 0.71, highlighting its ability to balance sensitivity (76%) and specificity (66%). These metrics underscore the robustness of the model in reliably identifying both high-risk and low-risk patients. The balanced performance across these metrics validates our model as a valuable tool for clinical decision support, enabling more informed and effective interventions in fall risk management. The model’s performance demonstrates the effectiveness of combining advanced machine learning techniques with electronic health records.

4.2 SHAP XAI Analysis

Figure 2 provides a global feature importance analysis using SHAP values, uncovering the most significant factors influencing fall risk predictions. The analysis identifies ethnicity (Non-Hispanic and

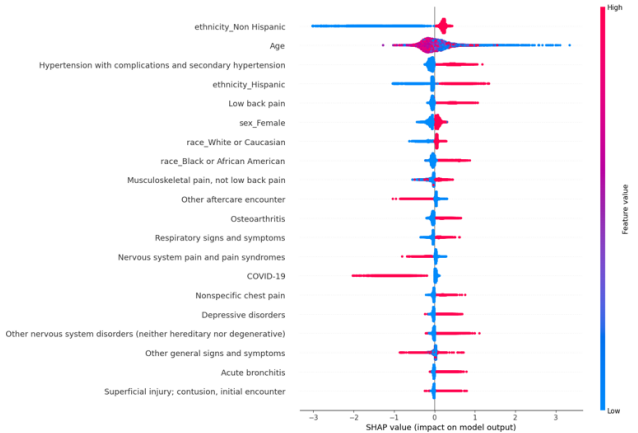


Figure 2: SHAP summary plot for the XGBoost model, showing the impact of each feature on the fall risk prediction. Features are ranked in order of importance, with higher-ranking features contributing more to predictions.

Hispanic), age, and hypertension with complications as the most impactful contributors, reflecting well-established fall risk factors in clinical studies. The important conditions such as low back pain, musculoskeletal pain, and osteoarthritis highlight the critical role of mobility-related impairments in increasing fall susceptibility. The analysis revealed patterns in feature importance, with certain health indicators showing stronger correlations with fall risk across the patient population. The SHAP framework enabled us to generate detailed explanations for each prediction, helping healthcare providers understand exactly which factors contributed to a patient’s risk classification.

Figure 3 illustrates a local SHAP analysis for individual patient predictions, highlighting how specific features contribute to moving the model output from the base value to the final prediction. The waterfall plots provide an in-depth view of the feature-level influences, where risk factors such as schizophrenia spectrum disorders, neurocognitive disorders, and hypertension with complications push the prediction higher, while protective factors such as younger age or the absence of respiratory symptoms decrease the predicted fall risk. These plots offer valuable insights into the personalized predictions, enabling clinicians to understand the driving factors behind the fall risk assessment for each patient.

To demonstrate the detailed feature-level analysis our system provides, we present a specific case study where SHAP values were

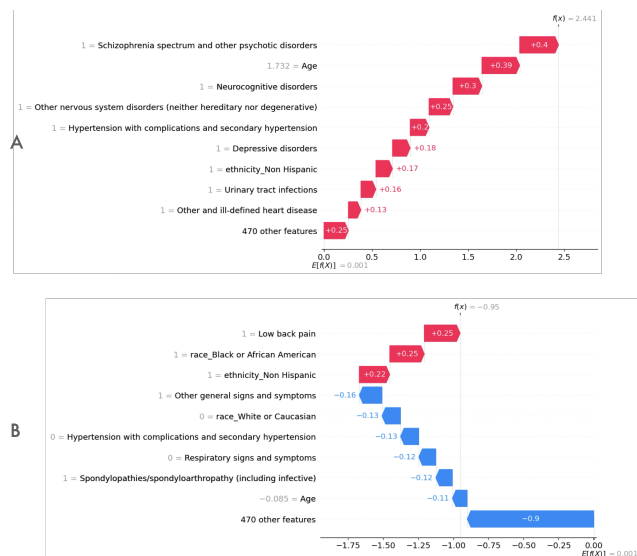


Figure 3: (A) SHAP waterfall plot for a high-risk patient, demonstrating the positive contributions of features such as schizophrenia spectrum disorders, neurocognitive disorders, hypertension with complications, and age toward a high fall risk prediction. The cumulative SHAP values highlight how each feature adds to the predicted fall risk score. (B) SHAP waterfall plot for a low-risk patient, showing the negative contributions of features such as younger age, absence of respiratory symptoms, and absence of hypertension with complications, reducing the overall fall risk prediction. The plot illustrates the balancing effect of protective factors against risk factors in determining fall risk.

used to explain an individual patient’s fall risk prediction (Figure 4). This detailed breakdown illustrates how our model evaluates multiple factors simultaneously, with some features increasing the risk (positive SHAP values) and others decreasing it (negative SHAP values). In this case, hypertension with complications and secondary hypertension emerged as the strongest risk factor, followed by conditions like other nervous system disorders, noninfectious gastroenteritis, and osteoarthritis. Demographic factors, such as being Non-Hispanic, also significantly contributed to the predicted fall risk. Interestingly, certain conditions, such as the absence of musculoskeletal pain, showed a protective effect, highlighting the nuanced interactions between risk factors.

The feature importance analysis was particularly insightful in identifying unexpected relationships and potential risk contributors, such as the positive impact of urinary tract infections and respiratory signs and symptoms. By examining SHAP values across patient subgroups, we could uncover patterns specific to certain clinical or demographic profiles. This granular understanding of feature interactions enhanced the model’s clinical relevance, enabling more personalized and actionable risk assessments and intervention strategies.

4.3 LLM-based Risk Interpretation

The integration of natural language generation capabilities through Langchain significantly enhanced the accessibility and utility of our model’s outputs. By transforming technical SHAP values and risk predictions into coherent, narrative explanations, we created an intuitive interface between the AI system and its clinical users. These generated narratives bridged the gap between complex mathematical models and practical clinical application. Our natural language explanations combined both high-level risk summaries and detailed feature-specific insights, providing a comprehensive yet accessible view of each patient’s risk profile. The narratives were structured to present information in a clinically relevant format, starting with the overall risk assessment and then systematically explaining the contributing factors. This approach ensured that providers could quickly grasp the essential information while having access to deeper insights when needed. To illustrate the system’s capability in generating clear and actionable explanations, we present three representative examples of the generated narratives, each demonstrating different risk levels and factor combinations (Figure 5).

The examples in Figure 5 demonstrate the system’s ability to generate comprehensive yet accessible explanations that effectively communicate risk levels, contributing factors, and their interactions. Each narrative maintains a clear structure, beginning with the overall risk assessment, followed by detailed factor analysis, and concluding with evidenced-based XAI recommendations [28]. The explanations successfully balance technical accuracy with accessibility, making complex medical predictions understandable to healthcare providers while maintaining clinical relevance.

The effectiveness of our methods in translating complex model outputs into comprehensible narratives marks a substantial improvement in making AI-driven healthcare tools more accessible to practitioners. The system’s ability to maintain clinical accuracy while presenting information in an intuitive format demonstrates the potential for broader adoption of AI-based decision support tools in healthcare settings.

5 Conclusion

This study presents advancement in fall risk prediction and clinical decision support through the novel integration of machine learning, explainable AI, and large language models. Our approach achieved 71% accuracy in fall risk prediction while maintaining high levels of interpretability through SHAP-based feature analysis and natural language explanations. The successful combination of XGBoost’s predictive power with sophisticated explainability techniques addresses a critical gap in clinical implementation of AI systems, where both accuracy and interpretability are essential for practical adoption. The integration of large language models for generating natural language explanations represents a particularly innovative contribution, transforming complex model outputs into actionable clinical insights. This advancement demonstrates how modern AI techniques can be leveraged to not only predict health risks but also communicate these predictions effectively to healthcare providers. The system’s ability to provide multi-layered explanations, from high-level risk assessments to detailed feature analyses, makes it a versatile tool for different clinical scenarios and

Risk Assessment: The patient was identified as having an 86.6% chance of falling in the future, indicating a high-risk level.

Detailed Feature Impact Analysis:

1.Primary Risk-Increasing Factors:

- **Hypertension with complications and secondary hypertension** (Impact: +0.4539, Present: Yes)
- **Other nervous system disorders (neither hereditary nor degenerative)** (Impact: +0.3537, Present: Yes)
- **Noninfectious gastroenteritis** (Impact: +0.2604, Present: Yes)
- **Osteoarthritis** (Impact: +0.2106, Present: Yes)
- **Urinary tract infections** (Impact: +0.1810, Present: Yes)
- **Ethnicity: Non-Hispanic** (Impact: +0.1627, Present: Yes)
- **Respiratory signs and symptoms** (Impact: +0.1454, Present: Yes)

2.Risk-Decreasing Factors:

- **Age** (Impact: -0.2229, Present: 1.0230123690638282)
- **Musculoskeletal pain, not low back pain** (Impact: -0.1803, Present: No)
- **Dysphagia** (Impact: -0.1564, Present: Yes)

Figure 4: Example case of fall risk explanation using SHAP XAI.

Example 1 (High Risk - Multiple Chronic Conditions):

"Based on the patient's medical history, the model has predicted that there is a 91.8% chance that the patient will experience a fall in the future, indicating a high risk. The patient's age, presence of schizophrenia and other psychotic disorders, hypertension with complications, neurocognitive disorders, other nervous system disorders, gender (female), ethnicity (non-Hispanic), urinary tract infections, depressive disorders, and other heart diseases are all factors that contribute to this high risk. "

Example 2 (Medium Risk - Complex Demographic and Medical Factors):

"Based on the model's predictions and explanations for this patient, it appears that there is a 58.2% chance that they may experience a fall in the future. This indicates a medium level of risk. The patient's neurocognitive disorders and being Black or African American contribute positively to this prediction. However, being White or Caucasian, and having no hypertension or respiratory signs and symptoms all have a negative impact on the prediction. Being female and having a history of other aftercare encounters slightly increase the risk. Overall, the patient's demographic and medical history play a significant role in their risk for falls. "

Example 3 (High Risk - Age and Medical History Interaction):

"Based on the model's predictions, the patient has a 79.8% chance of falling in the future, indicating a high risk. This is influenced by several factors such as age, ethnicity, and medical conditions. The patient's age has a positive impact on the prediction, meaning that the older the patient is, the higher their risk of falling. Being non-Hispanic and having a history of pneumonia and sinusitis also contribute to the risk. Other factors such as race, back pain, and genitourinary issues also have an impact on the prediction"

Figure 5: Fall risk interpretation using large language models

user needs. This work establishes a framework for developing more sophisticated and user-friendly clinical decision support systems.

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